

JEE MAIN REVISION MASTER: MATRICES AND DETERMINANTS SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 368 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Properties of Determinants & System of Linear Equations	272 PYQs	Determinants & Linear Systems
Very High	Matrix Inversion, Adjoint & Powers	98 PYQs	Matrix Algebra

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Properties of Determinants & System of Linear Equations	$\det(kA) = k^n \det(A)$, $\text{Unique: } \Delta \neq 0$, $\text{Infinite: } \Delta = \Delta_x = \Delta_y = \Delta_z = 0$, $\text{No sol: } \Delta = 0 \text{ \& at least one } \Delta_i \neq 0$
Matrix Inversion, Adjoint & Powers	$A \cdot \text{adj}(A) = (\det A) I_n$, $\det(\text{adj}(A)) = (\det A)^{n-1}$, $\text{adj}(\text{adj}(A)) = (\det A)^{n-2} A$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Properties of Determinants & System of Linear Equations (Very High Yield — 272 PYQs)

Core Principle: Evaluation of determinants, Cramer's rule, and consistency conditions for non-homogeneous and homogeneous linear systems.

■■ **Caution / JEE Pitfall:** For an $n \times n$ matrix A , $\det(kA) = k^n \det(A)$, NOT $k \det(A)$!

■ **Speed Trick:** A homogeneous system $AX = 0$ has non-trivial (infinite) solutions if and only if $\det(A) = 0$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "If the system of linear equations : $x+y+z=6$ $x+2y+5z=10$ $2x+3y+\lambda z=\mu$ has infinitely many s..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "If $f:N \rightarrow Z$ is defined by $f(n)=\begin{aligned} &n \& -1 \& -5 \parallel -2n^2 \& 3(2k+1) \& 2k+1 \parallel -3n^3 \& 3k(2k+1) \& 3k(k+2)+1 \end{aligned}$..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "If the system of equations : $x+y+z=5$ $x+2y+3z=9$ $x+3y+\lambda z=\mu$ has infinitely many solutions,..." [Ans: N/A]

3.2 Matrix Inversion, Adjoint & Powers (Very High Yield — 98 PYQs)

Core Principle: Properties of adjoint and inverse matrices, matrix multiplication transpose $(AB)^T = B^T A^T$, and symmetric/skew-symmetric decomposition.

■ **Caution / JEE Pitfall:** A skew-symmetric matrix of ODD order has determinant equal to ZERO ($\det(A) = 0$).

■ **Speed Trick:** For a 3×3 matrix: $\det(\text{adj}(A)) = (\det A)^2$, and $\det(\text{adj}(\text{adj}(A))) = (\det A)^4$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Let $P=[p_{ij}]$ and $Q=[q_{ij}]$ be two square matrices of order 3 such that $q_{ij}=2^{i+j-1}p_{ij}$ and $\det(Q)=2^{10}$. Then the value of $\det(P)$ is _____" [Ans: N/A]
- **[2026 Morning Shift] Q#1 (Numerical):** "Let $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ satisfy $A^2 + \alpha(\text{adj}(\text{adj}(A))) + \beta(\text{adj}(A)(\text{adj}(\text{adj}(A)))) = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ for some $\alpha, \beta \in \mathbb{R}$. Then $(\alpha - \beta)^2$ is equal to _____" [Ans: 4]
- **[2026 Evening Shift] Q#1 (Numerical):** "Let $A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{bmatrix}$ and B be a matrix such that $B(I-A) = I+A$. Then the sum of all elements of B is _____" [Ans: 3]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Let $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ satisfy $A^2 + \alpha(\text{adj}(\text{adj}(A))) + \beta(\text{adj}(A)(\text{adj}(\text{adj}(A)))) = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ for some $\alpha, \beta \in \mathbb{R}$. Then $(\alpha - \beta)^2$ is equal to _____

Correct Answer: 4

Solution Analysis: We need to find α and β from $A^2 + \alpha(\text{adj}(\text{adj}(A))) + \beta(\text{adj}(A)(\text{adj}(\text{adj}(A)))) = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$. Let us solve step by step. 1. Given matrix $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ 2. Find $\det(A)$ Since the third row is $(0, 1, 0)$, expand along the third row: $\det(A) = 1 \cdot \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = (-1)(0) - (1)(1) = -1$. So, $\det(A) = -1 \neq 0$. Hence A is invertible. 3. Use NCERT property of adjoint of adjoint For an $n \times n$ invertible matrix, $\text{adj}(\text{adj}(A)) = (\det A)^{n-2} A$. Here $n=3$, so $\text{adj}(\text{adj}(A)) = (-1)^{3-2} A = -A$.

Benchmark Problem 2: JEE Main 2026 (Online) 23rd January Morning Shift (Q#2)

Let $|A|=6$, where A is a 3×3 matrix. If $|\text{adj}(3\text{adj}(A^2 \cdot \text{adj}(2A)))| = 2^m \cdot 3^n$, $m, n \in \mathbb{N}$, then $m+n$ is equal to _____.

Correct Answer: 62

Solution Analysis: Let $X = A^2 \cdot \text{adj}(2A)$ so the given expression is $|\text{adj}(3\text{adj}(X))|$. Since A is a 3×3 matrix and $|A|=6$: Step 1: Find $|A^2|$ $|A^2| = |A|^2 = 6^2 = 36$ Step 2: Find $|\text{adj}(2A)|$ First, $|2A| = 2^3 |A| = 8 \cdot 6 = 48$ For a 3×3 matrix M , $|\text{adj}(M)| = |M|^{3-1} = |M|^2$ So, $|\text{adj}(2A)| = |2A|^2 = 48^2$ Step 3: Find $|X|$ $|X| = |A^2| \cdot |\text{adj}(2A)| = 36 \cdot 48^2$ Now, $36 = 2^2 \cdot 3^2$, $48 = 2^4 \cdot 3$ So, $48^2 = 2^8 \cdot 3^2$ Hence, $|X| = (2^2 \cdot 3^2)(2^8 \cdot 3^2) = 2^{10} \cdot 3^4$ Step 4: Find $|\text{adj}(X)|$ First, $|\text{adj}(X)| = |X|^{3-1} = |X|^2 = (2^{10} \cdot 3^4)^2 = 2^{20} \cdot 3^8$